Tribhuvan University

Faculty of Humanities and Social Science

Prithvi Narayan Campus, Pokhara

Lab Report on Computer Networking (CACS303)

BCA 5th Sem

[2079 Batch]

Submitted By

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Batch: 2079

Submitted To

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BCA Program, PNC

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S.N.	Objective	Date of submission	Remarks
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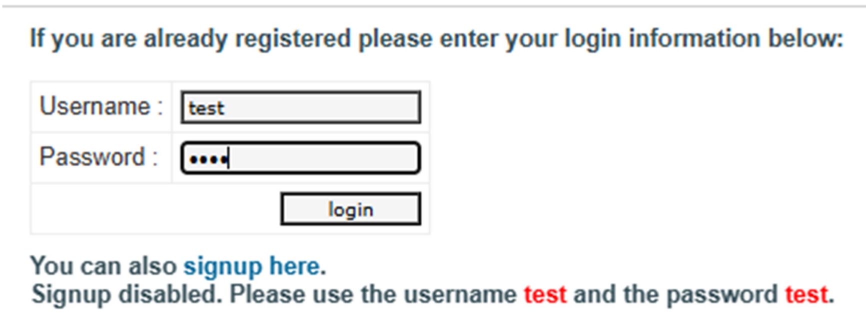
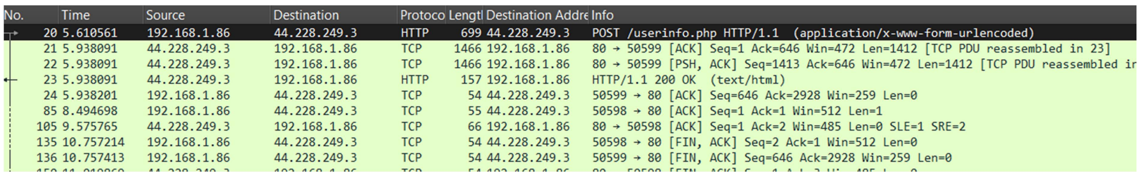
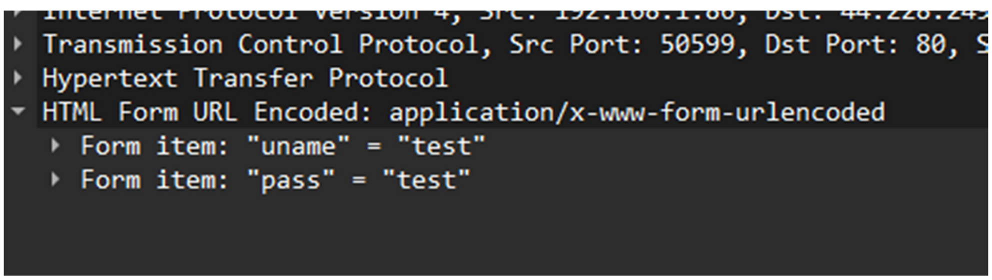
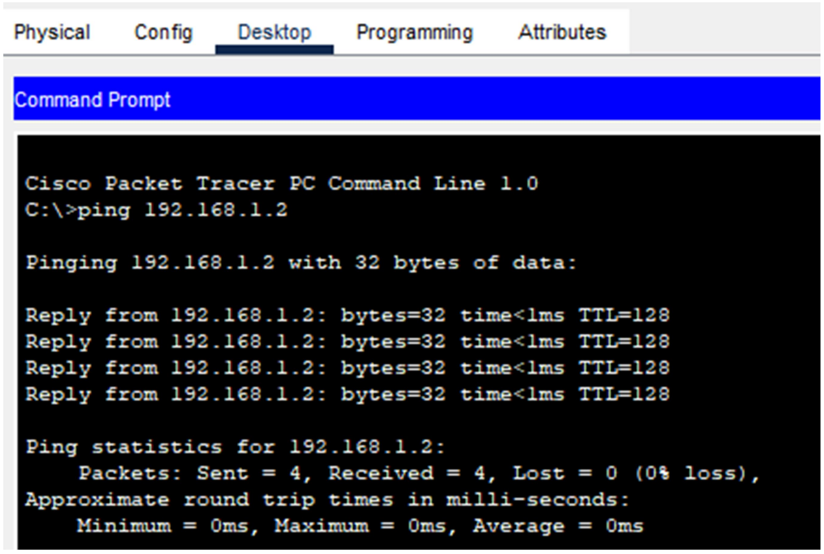


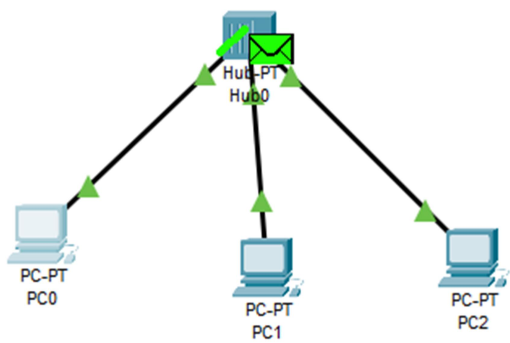


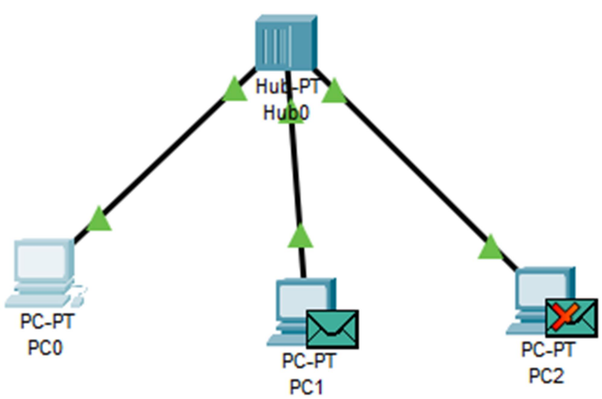


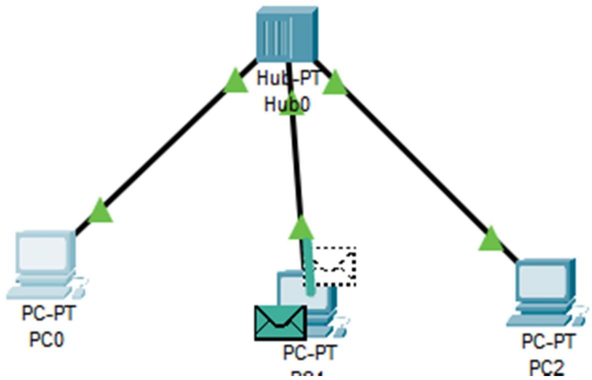
Conclusion

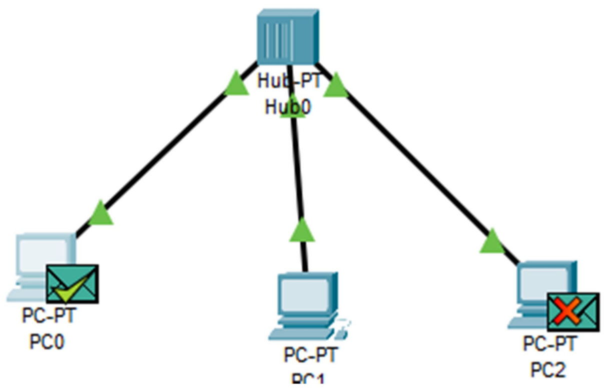
This lab demonstrates the security risk of using unencrypted protocols for transmitting sensitive information. Attackers on the same network can easily intercept login credentials if encryption is not used.


Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2
Pinging 192.168.1.2 with 32 bytes of data:
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms


Hub-PT
Hub0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2


Hub-PT
Hub0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2

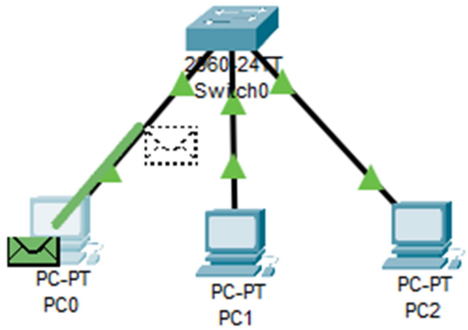




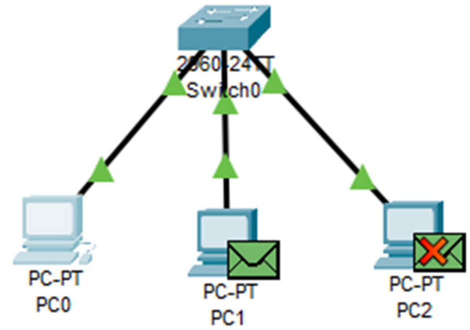
Conclusion

The Hub, a Layer 1 device, operates by broadcasting signals to all ports, leading to increased traffic and possible collisions. This behavior was clearly demonstrated using Cisco Packet Tracer, helping visualize the limitations and working mechanism of hubs in real-time.

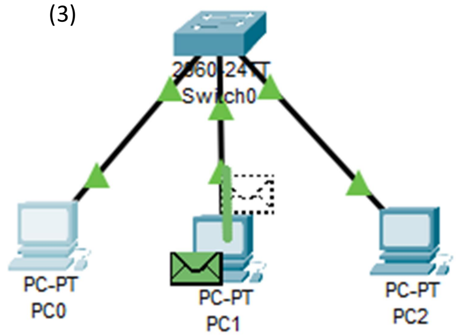
(1)



(2)



(3)



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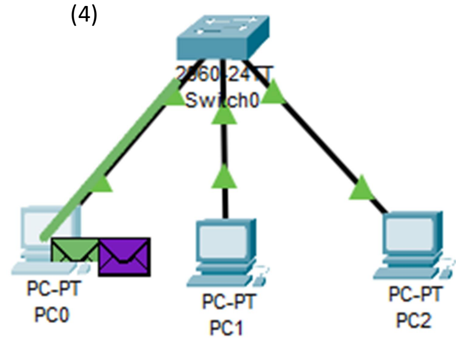

Conclusion

This lab demonstrated that a switch intelligently forwards traffic to the correct destination using MAC address tables. Unlike a hub, it reduces collisions and increases efficiency, making it ideal for modern LANs.